S.B. JAIN INSTITUTE OF TECHNOLOGY MANAGEMENT & RESEARCH, NAGPUR

Practical 11 Postlab (OPEN ENDED PRACTICAL)

Aim: To simulate a demand paging based virtual memory system in which pages are brought into main memory only when referenced, ensuring efficient utilization of memory resources.

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Aim: To study the working of demand paging mechanism and implement a system where memory allocation is done dynamically based on page requests.

❖ Objectives:

i. To understand on-demand memory allocation.
ii. To observe the behavior of page faults during execution.
iii. To evaluate memory performance using paging techniques.

Requirements:

✓ Hardware Requirements:

Processor: Minimum 1 GHz
RAM: 512 MB or higher
Storage: 100 MB free space

✓ Software Requirements:

Operating System: Linux/Unix-based
Shell: Bash 4.0 or higher
Text Editor: Nano, Vim, or any preferred editor

* Theory:

In modern operating systems, efficient memory management is essential for smooth execution of multiple processes. Demand paging is one such technique that allows a system to load only the required portions (pages) of a program into memory instead of loading the complete program at once.

Initially, all pages reside in secondary storage. When a process starts execution and attempts to access a page that is not present in RAM, the system triggers a page fault. The operating system then pauses the process, fetches the required page from disk, loads it into an available frame, and updates the page table.

This method reduces initial loading time and conserves memory, allowing more programs to run simultaneously. However, if page faults occur too frequently, it can lead to performance issues due to repeated disk access.

To manage this, the operating system uses page replacement techniques to decide which page should be removed when memory is full. The efficiency of demand paging largely depends on minimizing page faults and optimizing page replacement decisions.

Thus, demand paging provides a practical balance between memory usage and system performance

❖ CODE:

```
#include<stdio.h>

int main() {
    int pages[12] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int frames[3];
    int i, j, pointer = 0, flag, faults = 0, hits = 0;

    for(i=0;i<3;i++)
       frames[i] = -1;

    for(i=0;i<12;i++) {
        flag = 0;

        for(j=0;j<3;j++) {
            if(frames[j] == pages[i]) {
                flag = 1;
               hits++;
               break;
            }
        }

        if(flag == 0) {
            frames[pointer] = pages[i];

            pointer = (pointer + 1) % 3;

            faults++;

        }

    }
```

```
    printf("Total Hits = %d\n", hits);

    printf("Total Faults = %d\n", faults);


    return 0;

}
```

❖ Output:

input

Total Hits = 3
Total Faults = 9

...Program finished with exit code 0 Press ENTER to exit console.

Conclusion The implementation shows that demand paging helps in reducing memory usage by loading pages only when required. It improves system efficiency and supports execution of large programs. However, performance depends on reducing page faults through proper memory management strategies.

❖Discussion Questions:

1. What is the main purpose of demand paging?

2. What happens when a page is not available in memory?

3. Why is demand paging important in modern OS?

4. What is a page frame?

5. What is the disadvantage of frequent page faults?

References:

• Operating System Concepts Book
• https://www.tutorialspoint.com/

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Signature
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